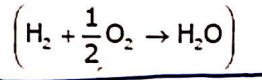


❖ **Example Calculation of Thermochemical Equilibrium in a Rocket Chamber:**

The rocket combustion chamber operates at $p = 210 \text{ atm}$ and an oxidizer-to-fuel mass ratio $O/F = 6$.

→ fuel rich

The stoichiometric ratio for full combustion



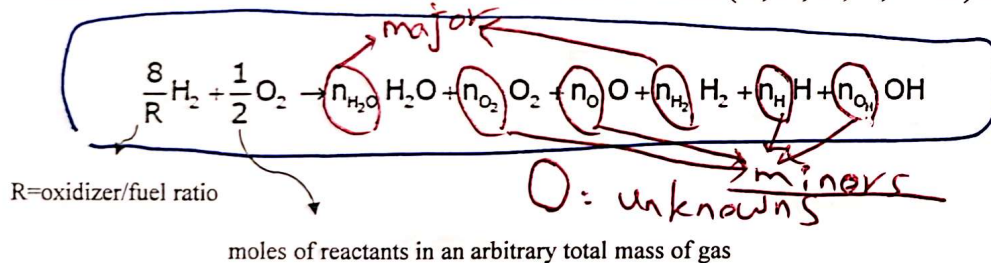
$O/F = 8$, → slightly fuel-rich and very high combustion temperature

I. Atomic Species Concentration

$$\text{st } R = r_{A=O} F = \frac{\text{mass of } \text{O}_2}{\text{fuel}} = \frac{\frac{1}{2} \times 32}{2 \times 1} = 8$$

Combustion of hydrogen in oxygen, with unknown amounts of water.

✓ The number of atomic moles for each kind of atom (O, H, N, C,) is the same before and after reaction



real reaction

$$\frac{8}{R} \text{H}_2 \text{ if } R = R_{\text{st}} \rightarrow \frac{8}{R} = 1$$

Here $R = 6$

$$N_{\text{O}_2} = 1/2$$

arbitrarily select

$$\frac{32 \times 1/2}{2N_{\text{H}_2}} = R$$

$$; \quad N_{\text{H}_2} = \frac{8}{R}$$

■ The list of candidate reaction products is arrived at from our background knowledge of what species actually do occur.

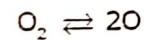
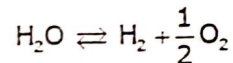
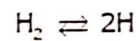
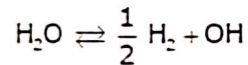
if R were low enough $\longrightarrow$ fuel-rich $\longrightarrow$ not include O₂ or O and oxygen species depleted.

- We would simply ignore them for practical purposes, as we ignore other exotic, but potentially real species, like , HO₂, H₃ , etc.

$$\begin{aligned} \text{H-conservation: } & \left\{ \begin{aligned} \frac{16}{R} &= 2n_{\text{H}_2\text{O}} + 2n_{\text{H}_2} + n_{\text{H}} + n_{\text{OH}} \\ \text{O-conservation: } & 1 = n_{\text{H}_2\text{O}} + 2n_{\text{O}_2} + n_{\text{O}} + n_{\text{OH}} \end{aligned} \right. \end{aligned}$$

Two equations involving the six unknown

The equilibrium conditions (laws of mass-action) for four elementary reactions selected so as to involve these decomposition products, as well as some of the "major" products.



equilibrium relationship

$$\frac{P_{\text{H}}^2}{P_{\text{H}_2}} = K_2(T)$$

$$\frac{P_{\text{H}_2} P_{\text{O}_2}^{1/2}}{P_{\text{H}_2\text{O}}} = K_3(T)$$

$$\frac{P_{\text{O}}^2}{P_{\text{O}_2}} = K_4(T)$$

$$P_{\text{OH}} = P \frac{n_{\text{OH}}}{n}$$

...

$$P_{\text{H}_2\text{O}} = P \frac{n_{\text{H}_2\text{O}}}{n}$$

$$(n = n_{\text{H}_2\text{O}} + n_{\text{O}_2} + n_{\text{H}_2} + n_{\text{H}} + n_{\text{OH}})$$

$$\frac{n_{\text{H}_2}^{1/2} n_{\text{OH}}}{n_{\text{H}_2\text{O}}} = \sqrt{\frac{n}{p}} K_1(T)$$

$$\frac{n_{\text{H}}^2}{n_{\text{H}_2}} = \frac{n}{p} K_2(T)$$

$$\frac{n_{\text{H}_2} n_{\text{O}_2}^{1/2}}{n_{\text{H}_2\text{O}}} = \sqrt{\frac{n}{p}} K_3(T)$$

$$\frac{n_{\text{O}}^2}{n_{\text{O}_2}} = \frac{n}{p} K_4(T)$$

rewrite (*) in terms of moles: $\frac{P_{\text{H}}^{1/2} P_{\text{OH}}}{P_{\text{H}_2\text{O}}} = \frac{(P)^{1/2} (\frac{n_{\text{H}}}{n})^{1/2} (P) \frac{n_{\text{OH}}}{n}}{(P) \frac{n_{\text{H}_2\text{O}}}{n}} = K_1(T)$

$$\Rightarrow \frac{P^{1/2} n_{\text{H}}^{1/2} n_{\text{OH}}}{n^{1/2} n_{\text{H}_2\text{O}}} = K_1(T) \Rightarrow \frac{P^{1/2} n_{\text{H}}^{1/2} n_{\text{OH}}}{n^{1/2} n_{\text{H}_2\text{O}}} = \sqrt{\frac{n}{p}} K_1(T)$$

$$K_1 = \exp \left(-\frac{\frac{1}{2} \mu_{H_2}^0 + \mu_{H_2O}^0 - \mu_{H_2O}^0}{RT} \right) \quad K_3 = \exp \left(-\frac{\mu_{H_2}^0 + \frac{1}{2} \mu_{O_2}^0 - \mu_{H_2O}^0}{RT} \right)$$

R=universal gas constant

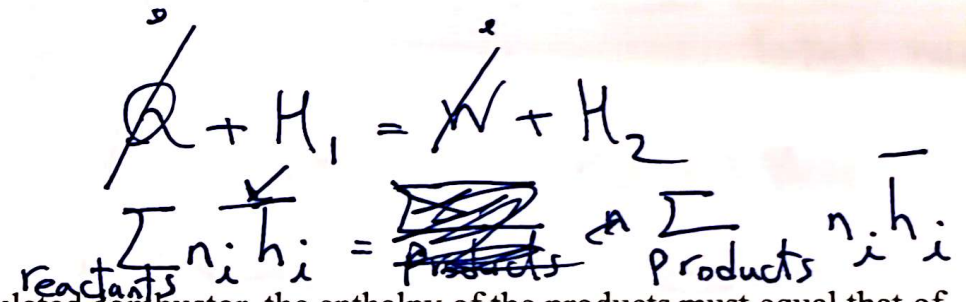
μ = chemical potential of species i at 1 atm. pressure, at the given temperature T

$$K_2 = \exp \left(-\frac{2\mu_{H_2}^0 - \mu_{H_2}^0}{RT} \right) \quad K_4 = \exp \left(-\frac{2\mu_{O_2}^0 - \mu_{H_2}^0}{RT} \right)$$

- The equilibrium constants still depend on temperature.

❖ Conservation of Energy

Since the reactions occur at constant pressure, if we assume a well-insulated combustor, the enthalpy of the products must equal that of the reactants.



$$\frac{8}{R} h_{H_2}(T_i) + \frac{1}{2} h_{O_2}(T_i) = n_{H_2O} h_{H_2O}(T) + n_{O_2} h_{O_2}(T) + n_O h_O(T)$$



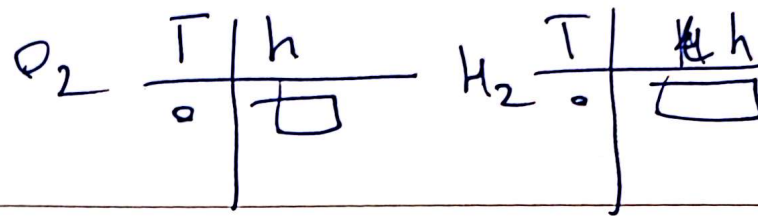
injection temperature

$$+ n_{H_2} h_{H_2}(T) + n_H h_H(T) + n_{OH} h_{OH}(T)$$

molar enthalpy of the species at temperature T

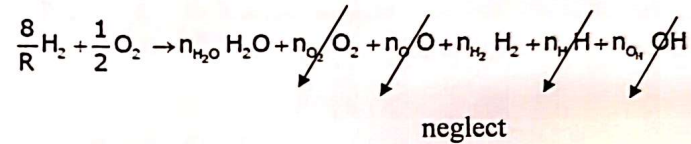


final equilibration temperature



❖ Numerical Iteration Procedure

R = O/F = 6, and P = 210 atm.



H-conservation: $\frac{16}{R} = 2n_{\text{H}_2\text{O}} + 2n_{\text{H}_2} + n_{\text{H}} + n_{\text{OH}}$

O-conservation: $1 = n_{\text{H}_2\text{O}} + 2n_{\text{O}_2} + n_{\text{O}} + n_{\text{OH}}$

minors

$$\frac{2n_{\text{H}_2\text{O}} + 2n_{\text{H}_2} = \frac{8}{3}}{n_{\text{H}_2\text{O}} = 1}$$

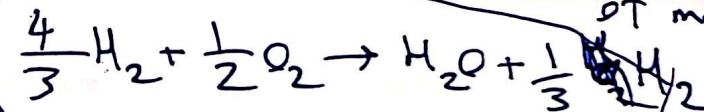
$$n_{\text{H}_2\text{O}} = 1 \quad n_{\text{H}_2} = 1/3$$

$$n = 4/3$$

total number of moles

The enthalpy before reaction (assuming very cold reactants $T_i = 0^\circ \text{K}$):

$$H = \frac{4}{3} h_{\text{H}_2}(\text{O}) + \frac{1}{2} h_{\text{O}_2}(\text{O}) = -15,632 \text{ J (for the } \frac{4}{3} \text{ moles of H}_2, \frac{1}{2} \text{ mole of O}_2 \text{).}$$



$$H_R = H_P$$

For the products to have the same enthalpy, the temperature must be very high. Using the table in Ref. 12.1 (pp. 692-693):

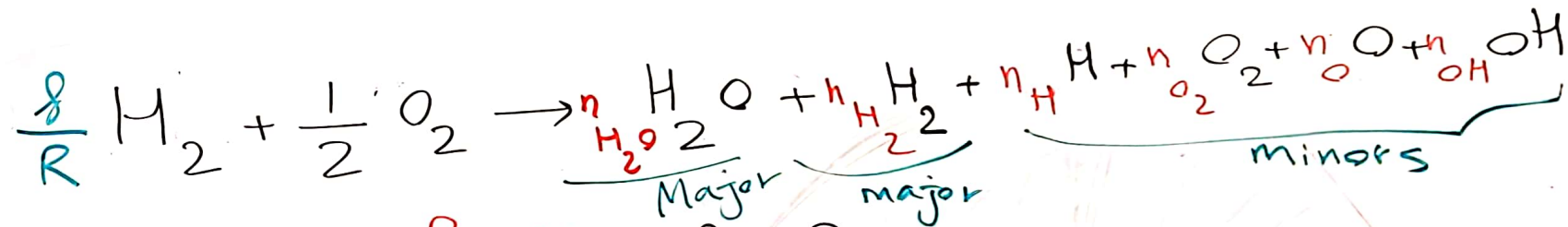
T(K)	3400	4000	4200
$h_{\text{H}_2\text{O}}$ (J/mole)	-92,973	-58,547	-46,924
h_{H_2} (J/mole)	103,738	126,846	134,700
$h_{\text{H}_2\text{O}} + \frac{1}{3} h_{\text{H}_2}$	-58,394	-16,265	-2,024

Linear interpolation between 4000 and 4200 K

$$-15,632 \text{ J}$$

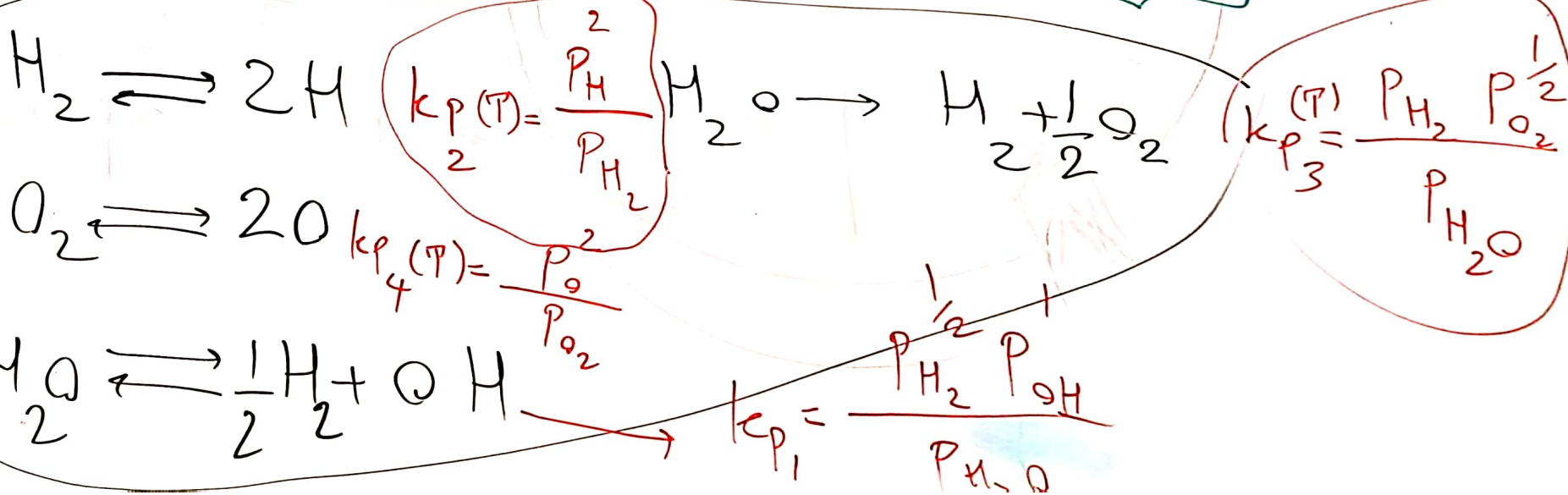
$$T = 4009 \text{ K}$$

<https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1&Type=JANAFG&Table=on>



Balance of H: $\frac{8}{R} \times 2 = 2n_{H_2O} + 2n_{H_2} + n_H + n_{OH}$

" " O: $1 = n_{H_2O} + 2n_{O_2} + n_O + n_{OH}$



ignore minors, obtain number of moles of products

Write Energy Eq. get T of products

Eqⁿ $k_p = k_p(T)$, read $k_p(T)$ for minor reactions

Solve for unknown numbers of $k_p(T)$ Eqⁿ

& get number of moles of minors

Correct the n of a majors

Write Energy Eq. again $\rightarrow$ get new T
 repeat until T does not change

$$X = \frac{P}{P_m} = \frac{n}{n_m} \Rightarrow P = \frac{P_m}{n_m} n \quad \text{replace this in } k_p(T)$$

$$n = n_m = n_{H_2O} + n_{H_2} + \cancel{n_{O_2} + n_{OH} + n_{O} + n_H}$$

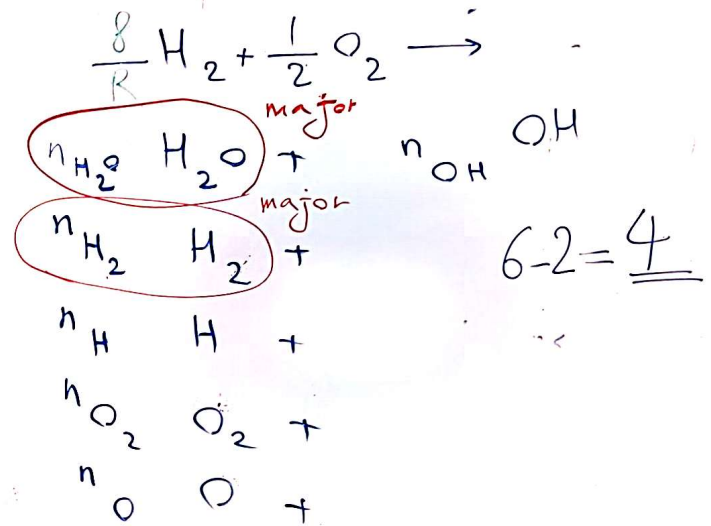
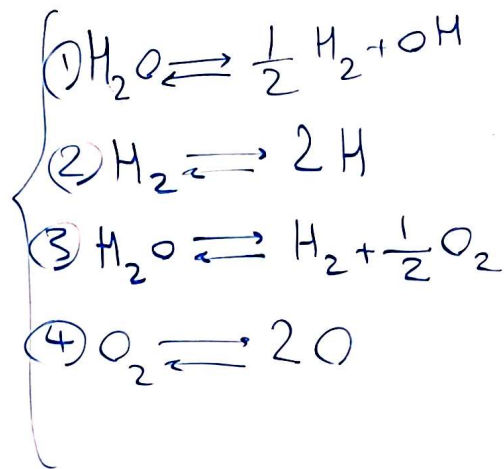
$P = P_m \rightarrow$ given by question

$$k_1(T) = \frac{P_{H_2}^{\frac{1}{2}} P_{OH}}{P_{H_2O}}$$

$$k_2(T) = \frac{P_H}{P_{H_2}}$$

$$k_3(T) = \frac{P_{H_2} P_{O_2}^{\frac{1}{2}}}{P_{H_2O}}$$

$$k_4(T) = \frac{P_O^2}{P_{O_2}}$$



$$\chi = \frac{P}{P_m} = \frac{n}{n_m} \quad (I) \xrightarrow{\text{put in } k_1(T)} \left(\frac{n_{H_2}^{\frac{1}{2}} n_{OH}}{n_{H_2O}} \right) = \sqrt{\frac{n_m}{P_m}} k_1(T) = \sqrt{\frac{n}{P}} k_1(T)$$

known:

$$n_m = (n_{H_2O} + n_{H_2} + n_H + n_{O_2} + n_O + n_{OH})$$

$$P_{H_2}^{\frac{1}{2}} = \left(\frac{P_m}{n_m} \right)^{\frac{1}{2}} n_{H_2}^{\frac{1}{2}}$$

$$P_{OH} = \left(\frac{P_m}{n_m} \right) n_{OH}$$

$$P_{H_2O} = \left(\frac{P_m}{n_m} \right) n_{H_2O}$$